

Exam 2 Guide

See the Topics Schedule and Assessment sections in the Syllabus for exam schedules and procedures.

Topics

1. Basics

- **2nd-Order ODE forms.** Constant and non-constant coefficients
- **Types of solutions.** General, specific, homogeneous, and particular
- **Verifying solutions.** 2nd-order ODEs

2. Modelling Physical Systems

- **Interpreting 2nd-order ODEs.** Making meaningful 2nd-Order linear ODEs
- **Spring-Mass.** Damped and undamped harmonic oscillators
- **Spring-Mass with external force.** Forced oscillations and resonance

3. Qualitative Analysis of 2nd-Order Linear ODEs

- **Existence and uniqueness.** Initial value problems and equilibriums
- **Equilibrium solutions and stability.** Stable and unstable (nodes and spirals)

4. Solving 2nd-Order Linear ODEs

- **Homogeneous solutions.** Characteristic equations and its roots
- **Non-Homogeneous solutions.** Particular solutions using the following methods:
 - **Undetermined coefficients.** Create a trial solution and solve the coefficients
 - **Variation of parameters.** Using the homogeneous solutions and the Wronskian

5. Order Reduction

- **Substitution.** Reduce a 2nd-order ODE into 1st-order
- **Transformation.** Convert a linear 2nd-Order ODE to a linear system of 1st-order ODEs